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**CAPTURE OF FOCAL MULTIDIMENSIONAL
DATA IN HUMAN AND LLM-BASED FACT-
CHECK SUMMARIES**
BACHELOR THESIS

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Abstract

In the contemporary information landscape, marked by a surge in data and the rise of "post-information age" challenges, fact-checking journalism has become essential to combat the spread of disinformation by contextualizing public claims. As the volume of fact-checking articles grows, effective summarization is crucial to make these detailed analyses accessible. However, this presents a significant problem: both human and automated summaries risk omitting critical, multidimensional data—such as time, location, and methodology—thereby undermining the factual grounding and value of the summary.

This thesis addresses this challenge by first defining "focal multidimensional data" within fact-checking articles. It then develops a novel suite of custom metrics designed to measure the capture of this data, including Data Point Coverage, Dimensional Completeness, and Weighted Data Point Coverage. This framework was applied to a curated dataset of 14 climate-related fact-checks from PolitiFact and their corresponding summaries generated by the gpt-4o-mini large language model (LLM).

Our findings indicate that while LLM-generated summaries achieve high semantic coherence (0.86 BERTScore F1), they fail to preserve the underlying evidence, capturing only 37% of the most critical, weighted data points. The analysis further reveals that while dimensions like subject and source are well-preserved, summaries frequently omit the precise quantitative value and, most critically, the temporal context of the data.

This study concludes that standard summarization metrics are insufficient for evaluating content in high-stakes, factual domains. The significant gap between semantic fluency and factual integrity highlights the urgent need for dimension-aware evaluation and the development of summarization models explicitly optimized to preserve the dimensional context that underpins factual claims.

Keywords

Fact-Checking, Text Summarization, Large Language Models (LLMs), Multidimensional, Data, Evaluation Metrics, Factual Integrity, Information Loss, Data Dimension, Quantity Extraction

Abstrakt

V současném informačním prostředí, které se vyznačuje prudkým nárůstem dat a vzestupem výzev „post informačního věku“, se stalo ověřování faktů v žurnalistice zásadní pro boj proti šíření dezinformací prostřednictvím kontextualizace veřejných tvrzení. S rostoucím objemem článků ověřujících fakta je pro zpřístupnění těchto podrobných analýz zásadní efektivní sumarizace. To však představuje významný problém: jak u lidských, tak u automatizovaných shrnutí hrozí, že budou vynechány kritické, vícerozměrné údaje - například čas, místo a metodika -, čímž se podkopává faktická podloženost a hodnota shrnutí.

Tato práce se zabývá tímto problémem tím, že nejprve definuje „stěžejní vícerozměrné údaje“ v rámci článků ověřujících fakta. Poté vyvíjí nový soubor vlastních metrik určených k měření zachycení těchto údajů, včetně pokrytí datových bodů, úplnosti dimenzí a váženého pokrytí datových bodů. Tento rámec byl aplikován na kurátorovanou sadu dat 14 faktů souvisejících s klimatem z PolitiFact a jejich odpovídající shrnutí vygenerovaná pomocí velkého jazykového modelu (LLM) gpt-4o-mini.

Naše zjištění ukazují, že souhrny vytvořené pomocí LLM sice dosahují vysoké sémantické koherence (0,86 BERTScore F1), ale nedokážou zachovat základní důkazy a zachycují pouze 37 % nejkritičtějších, vážených datových bodů. Analýza dále ukazuje, že zatímco dimenze jako předmět a zdroj jsou dobře zachovány, souhrny často opomíjejí přesnou kvantitativní hodnotu a, což je nejkritičtější, časový kontext dat.

Tato studie dochází k závěru, že standardní metriky sumarizace jsou pro hodnocení obsahu ve faktografických doménách s vysokou mírou závažnosti nedostatečné. Značný rozdíl mezi sémantickou plynulostí a faktickou integritou zdůrazňuje naléhavou potřebu hodnocení zohledňujícího dimenze a vývoje sumarizačních modelů explicitně optimalizovaných pro zachování dimenzionálního kontextu, který je základem faktických tvrzení.

Klíčová slova

Fact-Checking, Sumarizace textu, Velké jazykové modely (LLM), Vícerozměrná data, Metriky hodnocení, Faktická integrita, Ztráta informací, Dimenze dat, Extrakce kvantit

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1 Introduction

The contemporary information landscape is characterized by an unprecedented volume of data, a phenomenon significantly accelerated by Artificial Intelligence (AI) advancements. This so-called "post-information age" has given rise to a significant challenge: the blurring of lines between objective truth and subjective opinion, the second being often presented as a verifiable fact. In this environment, the source and wider context of presented information have become paramount in establishing credibility. As a direct response to the proliferation of disinformation, fact-checking journalism has emerged as a crucial discipline, aiming to combat falsehoods by thoroughly contextualizing claims made in public discourse.

Initially focused on the statements of politicians, the practice of fact-checking has evolved to address a wide array of claims found across media and society. Organizations like PolitiFact (PolitiFact, n.d.) produce complex analytical articles that not only verify statements but also provide in-depth background information, culminating in a clear verdict. As the volume of this fact-checking content grows, so does the demand for effective summarization to make these detailed reports accessible to a wider audience with diverse information needs (Haniková et al., 2024).

Recent advances in Large Language Models (LLMs) have demonstrated promising capabilities in summarization tasks. Indeed, some research has found that in the context of general news, "LLM summaries are judged to be on par with human-written summaries" (Zhang et al., 2024). In its essence, text summarization is a task of distilling presented information from a source text into a shorter format (Radev, 2002). However, fact-checking articles present a unique challenge. They are not merely journalistic pieces but are analytical documents designed to deliver a specific, evidence-based verdict. The central challenge of summarizing these reports, therefore, is to ensure that the generated digest retains sufficient informational value for a reader to comprehend and trust the final ruling.

The factual basis of these articles often relies on focal data points sets of values that are meaningless without clearly defined attributes such as the time period of an observation, the geographical region, the method of data collection, and the expressed units. We refer to these attributes as the data's dimensions. The concept of treating textual information in such a structured, multidimensional manner has precedent in data systems research, which has explored frameworks for querying unstructured text using principles from Online Analytical Processing (OLAP) (Lin et al., 2008). In the context of fact-checking, this dimensional perspective is not merely an analytical convenience; it is essential. The veracity of a claim for instance, "emissions have decreased by 10%"—can be entirely altered if a single dimension, such as the time period or geographical region, is omitted or changed.

This leads to the central problem this thesis addresses: the potential for both human-written and LLM-automated summaries to omit these critical data dimensions, thereby stripping the summary of its factual grounding and potentially misrepresenting the original fact-check's conclusion. Standard summarization metrics, which measure fluency and word overlap, are blind to this loss of dimensional integrity.

To address this gap, this study proposes and evaluates a novel framework for quantifying the completeness of summaries of fact-checking reports, with a specific focus on the capture of multidimensional data. The research is based upon previous work conducted within the CIMPLE project (CIMPLE | CHIST-ERA, 2020) which focuses primarily on climate-related fact-checking articles.

This thesis seeks to answer the following research questions:

RQ1: To what extent do LLM-generated summaries of fact-checking articles capture the focal multidimensional data present in the source text?

RQ2: What specific types of data dimensions (e.g., temporal, geographical, source) are most frequently lost or altered in the summarization process?

RQ3: How do standard, semantics-based evaluation metrics (like BERTScore) compare to custom, dimension-aware metrics in assessing the factual integrity of a fact-check summary?

To answer these questions, this thesis will first establish a theoretical background on fact-checking and summarization evaluation. It will then detail the methodology for the manual annotation of multidimensional data and the proposal of custom evaluation metrics. Subsequently, the results of applying this framework to a curated dataset will be presented and analyzed. Finally, the discussion will interpret these findings and consider their implications for the future of automated summarization in high-stakes, factual domains.

2 Related work

The following is an overview of the literature written on topics related to the proposed task. This part includes mention of recent switch from fact-checking on certain social media, extraction of entities, differently structured data, and multidimensionality with respect to textual data. The chapter also covers fundamental metrics used in the analysis of summaries.

2.1 The fact-checking landscape

The fact-checking phenomenon gained popularity on social networks. Meta has employed several non-government organizations, such as PolitiFact, to expertly identify social media posts and check the claims they present. Recently, Meta has stopped the service and chosen the approach of X (Isaac & Schleifer, 2025). The Community Notes, based on a voting system of contributing users (X, n.d.) took the place of curated posts and their rulings. Nevertheless, for the fact-checking organizations, these activities were only a fraction of their services (Isaac & Schleifer, 2025). The produced articles are available, and their informational value has not changed. They are written by professionals and present a unique dataset for consideration.

2.2 Entity extraction

While Named Entity Recognition (NER) is a foundational task in Natural Language Processing (NLP), this thesis will concentrate primarily on the extraction of quantities. The focus is justified by the fact that quantities are at the center of the dimensional approach for structuring data. Göpfert et al. (2022) described quantity as a composition of numeric value, unit, and modifier altering the meaning of quantity, such as ‘average’, ‘approx.’, or ‘above’.

However, the task of obtaining quantities from text is far from trivial. As detailed by Göpfert et al. (2022), measurements are expressed in a myriad of complex ways that extend beyond simple value unit pairs. For instance, numeric values can be expressed in digits (1, 2, 3...), words (one million), or fractions. Furthermore, a common linguistic pattern is coordinated listing of values where the order is critical: “*Pizza and Coffee cost 4 and 5 Euro, respectively*”.

The inherent difficulty of Quantity Extraction is significant for the proposed methodology in this thesis. Being able to reliably identify complex and heterogeneous quantitative representations is the necessary first step for analyzing multidimensional factual data.

2.3 Unstructured data

The fact-checking articles are inherently unstructured documents. Unlike common well-organized data meant for storage, sharing, or computer interaction, text of natural language lacks a formal data model. Predefined table-like structures are nowhere to be found in a prosaic narrative from humans made to be read by humans.

The heterogeneous character of the articles underlines the necessity for their in-depth study. Their structure is often highly flexible, with each article being unique due to the content it is based upon, the author's experience, and skill. Fundamentally, however, these articles are grounded in facts and data. The data journalism introduces a challenge since it is not a strictly number-oriented format, neither it is a literary art. Despite its difficult nature, this dataset is selected in this thesis for its unique informational value and cultural significance. The methodology of this thesis cannot be easily automated precisely due to the unstructuredness; therefore, it will incorporate manual methods and verification.

2.4 Multidimensionality

A raw numerical value isolated from its context is little more than a point in a void. What makes it valuable is the definition of its aspects and attributes. By grounding the point by dimensions transforms it into meaningful information. What does the data measure? During what time period was the observation done? What geographical area was considered? And according to what methodology? Answering these valid questions is needed in order to give the data its proper meaning and value.

The collective set of attributes defines the multidimensional space in which the data point is situated. The concept of multidimensionality is central to this work, especially as the domains of dimensions often overlap. For instance, a single data point measuring CO₂ emissions might have dimensions that capture the geographical aspect (e.g., a specific country), time period (a certain year), and methodology (self-reported government figures or remote sensory observations). The veracity of a claim is closely dependent on the precise definition of each consistent dimension.

The concept of dimensionality is widely used in business intelligence and the domain of data warehousing. Dimensions in the OLAP sense define the grain of the fact tables and tell the scope of measurement (Kimball, 2013). The numerical facts are stored within a data cube, allowing access to it by "slice and dice". The central measure (e.g., total sales) could be viewed by year, product category, or by store location (Kimball, 2013).

While fact-checking articles are not strictly structured database entities, this thesis argues that the data points (measurements) can be analyzed through a similar multidimensional lens.

2.5 Text Cubes

The principle of applying multidimensional analysis to unstructured text is not merely in theoretical level. It has been established domain of research in data systems. Researchers have developed sophisticated frameworks to transform text corpora into query-able, cube-like structures. Key examples include TextCube (Lin et al., 2008), which integrates Information Retrieval measures like Term Frequency (TF-IDF) into the cube to allow for keyword-based analysis across dimensions. The creation of such structures can be automated making it efficient even for massive text documents (Han, 2019) even in unsupervised way (Tao et al., 2018). TopicCube (Zhang et al., 2009) by contrast uses probabilistic topic modeling to organize documents. This enables to query the system based on semantic meaning rather than specific words. More advanced systems like EventCube (Tao et al., 2013) combine both of these methods and further enrich the data by extracting named entities (like people, organizations, and locations) to map their relationships in the database system.

2.6 Metrics

This subpart is dedicated to metrics used in the context of summarization. Quantification of summaries can for comparison of different authors, styles and express the qualities of different texts. Fabbri et al., (2021) in their meta study identified twelve main automatic evaluation metrics. The metrics can be divided into several categories based on their calculation: N-gram Overlap Metrics, Embedding-based Metrics, Model-based Metrics and Descriptive Statistical Metrics.

N-gram Overlap Metrics are based upon overlap of words or token sequences, where the length of adjacent subsets (grams) is the n (n-grams). This approach directly compares summary with a reference. The standard metrics for text summarization like ROUGE (Lin, 2004), which is recall-oriented and BLEU (Papineni et al., 2002), a precision-oriented metric often used in assessing translation are typical examples of this approach.

Embedding-based Metrics use word embeddings to compare the meaning of summarized text with a reference. The embedding aspect is a way of transforming text in a numerical representation. The representation is usually a multidimensional vector that can capture the meaning of words and the context in which they are used. Observing similarity in the high-dimensional space can capture the underlying nuances and implicit details of the summary. Metrics like BERT-score and MoverScore are prominent examples of this approach. They are computationally more expensive, but they often correlate better with human annotators (Zhang et al., 2019).

Model-based Metrics takes additional resources – a model, to evaluate the summarization of text. The model generates question-answer pairs from the full text to assess the integrity and the preservation of key facts in the summary. The agentic approach is a rather complex but shows significant correlation with human annotators (Wang et al., 2020).

3 Methodology

3.1 Dataset Curation

The initial corpus for this study consisted of fact-checking articles from PolitiFact related to the topics "Climate" and "Environment," which were identified via keyword search. The raw dataset used in this study was collected via web scraper by Mládek (2025) as part of a related thesis project.

To construct the final dataset, a two-pronged selection strategy was employed, combining purposive manual selection with a systematic, automated analysis. The first stage was a manual process aimed at capturing the heterogeneity of the source material. Eight articles were purposively chosen to form a diverse foundational set, selected for their representative mix of topics, writing styles, and the nature of their claims.

The second stage was to complement this manual selection by systematically identifying the remaining six articles. This was achieved using a custom script developed to score articles based on their density of quantitative data. The script was built using the SpaCy library (Honnibal et al., 2020) and employed a hybrid strategy combining rule-based matching with a pre-trained statistical model. The primary component was a rule-based system built with SpaCy's Matcher. A comprehensive lexicon of measurement units was compiled (e.g., "tons," "km," "acres," "years"). A specific pattern was then engineered to detect sequences where a number-like token was immediately followed by a unit from this lexicon.

This rule-based component was augmented by leveraging the pre-trained Named Entity Recognition capabilities of the "en_core_web_sm" pipeline (Honnibal et al., 2020). This allowed for the identification of built-in entity types such as *MONEY*, *PERCENT*, and *CARDINAL*. The script integrated the outputs from both the Matcher and the NER model, with a de-duplication step to ensure each measurement was counted only once.

This script was then run on the remaining articles in the corpus to assign each a quantitative score. To ensure the final selection represented a range of complexity, these scored articles were grouped into three strata-low, medium, and high quantity counts. Two representative articles were then selected from each stratum. This automated process added the final six systematically chosen articles to the corpus.

The combined result was a candidate pool of 14 articles. Each was manually verified to ensure its core arguments were fundamentally based on quantifiable evidence, constituting the final, curated dataset for this thesis.

3.2 Dimension Identification

The manual extraction of multidimensional data from the source articles was guided by a systematic, rule-based annotation protocol. This process, a form of manual content

analysis, ensures that the identification of dimensions is as consistent and reproducible as possible. The protocol was inspired by the principles of Quantity Extraction as described by Göpfert et al. (2022) and was conducted in a series of defined steps:

Step 1: Focal Point Identification (Quantity Extraction)

The initial step was to identify all focal data points within the text. A data point was defined as any mention of a quantity. The text was scanned for numerical values (e.g., "1.75 trillion," "555"), word-based quantities (e.g., "one million"), and fractions (e.g., "two-thirds"). For each identified quantity, the core measurement was recorded, consisting of its Value, its associated Unit (e.g., "USD," "Rank," "% of GDP"), and any explicit Modifier that altered its meaning (e.g., "approximately," "more than," "average").

Step 2: Contextual Dimension Annotation

For each focal point identified in Step 1, the surrounding text was analyzed to link it to its core contextual dimensions. Subject: The entity being measured was identified and recorded (e.g., "Build Back Better bill," "Firearm Deaths"). Time Period & Region: The sentence containing the focal point, as well as the immediate preceding and succeeding sentences, were scanned for explicit mentions of temporal or geographical context. If found, they were linked to the focal point.

Step 3: Provenance Annotation

To capture the evidentiary basis of the data, the source of each focal point was identified. Source (within text): Explicit attributions within the narrative (e.g., "according to a report by," "the study found") were recorded. Source (cites): The specific entity credited with the information (e.g., "Congressional Budget Office," "The Fact-Checker") was recorded.

Step 4: Rules for Ambiguous Cases

During the annotation process, several recurring ambiguities were resolved by applying a consistent set of rules:

- Rule for Unit Granularity: For derived metrics (e.g., "emissions per person"), the Unit was recorded as a standard unit in physics ("tons ") to preserve the representation from the source text yet remain comparable.
- Rule for Relative Values: Quantities expressed in relation to another variable (e.g., "twice as much as China") were recorded as proportional expressions to maintain their relational nature.
- Rule for Non-Specific Quantifiers: Vague qualifiers (e.g., "close to 500," "almost") were recorded in the Value dimension.
- Rule for Ranked Data: Enumerations or rankings (e.g., "France finished first...") were treated as a series of distinct data points, with the Value being the ordinal number and the Unit being "Rank."

While this protocol provides a systematic framework, it is important to note that its application by a single annotator is a limitation of this study. The primary goal of this

thesis is to explore the feasibility and utility of a dimension-aware evaluation. Therefore, this protocol serves as a proof-of-concept. A future validation of this methodology would require a study with multiple annotators and a calculation of inter-annotator agreement to establish its formal reliability.

3.3 Summarization

The summarization of fact-checking articles can be approached in several ways, including human-authored abstracts and machine-assisted methods. The PolitiFact articles in the dataset already contain human-written "If your time is short" bullet points. However, a preliminary analysis showed these often serve as high-level topic pointers or verdict statements rather than data-rich digests of the article's evidence.

Therefore, to directly test the preservation of focal multidimensional data from the full text, this study focuses on machine-generated summaries. Recent work suggests that LLMs tend to be more extractive when summarizing (Zhang et al., 2024), which forms a key hypothesis for this thesis: this extractive tendency might lead to a higher capture rate of factual data points compared to more abstractive human summaries.

To test this, an automated summary was generated for each of the 14 curated articles using the gpt-4o-mini (OpenAI, 2024) large language model. A simple, zero-shot prompting strategy was employed, instructing the model to "sum up the fact-checking article" without providing further constraints or examples. This baseline approach was chosen to test the model's inherent capabilities in this summarization task. These generated summaries serve as the primary subject for the quantitative analysis in this thesis.

3.4 Proposed Metrics

To comprehensively evaluate the quality of the generated summaries, a dual approach was adopted. First, standard, widely-used metrics were employed to establish a baseline for semantic and lexical quality. Second, a suite of custom metrics was designed specifically to address the primary research question of this thesis: the preservation of focal multidimensional data.

3.4.1 Standard Baseline Metrics

Standard metrics were used to characterize the summarization style of the LLM. For these calculations, the LLM-generated summary serves as the "candidate" text, and the full source article serves as the "reference" text. This approach does not measure quality against a golden summary, but rather quantifies the summary's faithfulness and overlap with the source material.

ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

This metric measures the lexical overlap between the summary and the source article. A high ROUGE score indicates a more extractive summary that reuses phrases from the source. The following variants were calculated:

- ROUGE-1: Overlap of individual words (unigrams).
- ROUGE-2: Overlap of two-word phrases (bigrams).
- ROUGE-L: Overlap based on the longest common subsequence, which captures sentence-level structure.

3.4.2 BERTScore

This metric moves beyond lexical overlap to measure semantic similarity. It uses contextual embeddings from BERT (Devlin et al., 2019) to assess whether the summary and the source text convey similar meanings, even if they use different words. The precision, recall, and F1 scores were calculated, with the F1 score being the primary measure of semantic faithfulness.

3.4.3 Proposed Custom Metrics for Dimensional Analysis

Given that standard metrics are blind to the structured, dimensional nature of factual data, a suite of custom metrics was developed. These metrics quantify the summary's ability to preserve the specific, verifiable evidence from the source text.

Data Point Coverage

This metric provides a raw measure of how many distinct facts from the source article were successfully included in the summary. Within the framework of this thesis, each "distinct fact" is an individually annotated quantity, complete with its unique dimensional context (such as its value, unit, subject, and time period). The metric is therefore calculated by dividing the number of unique quantities found in the summary by the total number found in the original source text.

$$\text{Coverage} = \frac{\text{Count of unique quantities captured in summary}}{\text{Total count of quantities points in source text}}, \quad (1)$$

Dimensional Completeness (Recall)

This metric assesses the depth of the captured information. For the data points that were included in the summary, it measures what fraction of their original dimensional context (e.g., Time, Region, Source) was also preserved. This answers the question: "Of the facts that were included, how complete are they?"

$$\text{Completeness} = \frac{\sum_{dp \in \text{Captured}} \text{Dimensions}_{\text{summary}}(dp)}{\sum_{dp \in \text{Captured}} \text{Dimensions}_{\text{source}}(dp)}, \quad (2)$$

Where dp is a data point (dimension of quantity), and Captured is the set of data points present in the summary.

Weighted Data Point Coverage

To assess a summary's ability to preserve the most salient information, the Weighted Data Point Coverage metric was developed. This metric is founded on the principle that not all facts within a fact-checking article are of equal importance. For a summary to be trustworthy, it must prioritize the data points most essential for a reader to comprehend and verify the article's final verdict. The metric refines the simple Data Point Coverage score by assigning a weight to each data point based on its proximity and criticality to the article's main claim. This creates a hierarchy of informational importance, defined as follows:

- **Weight of 5 (Primary Claim Data):** Assigned to data points directly mentioned in the fact-check's primary claim. The omission of this data fundamentally compromises the summary's core purpose.
- **Weight of 4 (Critical Evaluation Data):** Assigned to data points that are critical to the analytical process of evaluating the claim. This data constitutes the primary evidence supporting the verdict.
- **Weight of 3 (Source-Attributed Data):** Assigned to data points explicitly attributed to cited sources and experts. This information establishes the evidentiary authority of the fact-check.
- **Weight of 1-2 (Supporting Data):** Assigned to data points that provide additional context but are not central to the main argument. Their omission is less detrimental to the summary's factual integrity.

The final score measures the fraction of total "importance" captured by the summary.

$$\text{Weighted Coverage} = \frac{\sum_{dp \in \text{Captured}} \text{Weight}(dp)}{\sum_{dp \in \text{All}} \text{Weight}(dp)}, \quad (3)$$

Together, these three custom metrics provide a multi-faceted view of a summary's performance, moving beyond fluency to quantify its factual and dimensional integrity.

4 Results

This chapter presents the quantitative findings derived from the analysis of 14 curated fact-checking articles and their corresponding machine-generated summaries. Before presenting the numerical values, it is important to reiterate a key decision regarding the scope of this evaluation. While the source PolitiFact articles contain human-written "If your time is short" bullet points and "Our ruling", were deliberately excluded from the quantitative analysis.

A preliminary review, as detailed in the methodology, revealed that these human-authored abstracts primarily function as high-level topic pointers or verdict statements rather than as data-rich digests of the article's evidence. Because they are not designed to capture the specific, focal multidimensional data that is the subject of this research, they were deemed unsuitable for this study's evaluation framework. Therefore, the following results concern exclusively to the performance of the LLM-generated summaries in preserving the factual and dimensional integrity of the source text.

4.1 Summary of the Annotated Dataset

The manual annotation protocol was applied across the entire dataset, comprising 14 curated fact-checking articles. This process resulted in the identification of 206 unique multidimensional data points in total from the full source texts. The density of factual data varied significantly across the corpus, highlighting the heterogeneity of data journalism; the number of quantities in a single article ranged from zero ("Fighting cap and trade") to as high as 45 ("Texas is #1"). In the corresponding LLM-generated summaries for these articles, a total of 57 data points were captured. This annotated dataset forms the foundation for the detailed evaluation presented in the following sections.

ID	Articles	Full text	Summary	Ratio
1	Cutting Emissions	23	5	0,22
2	Billionaire couple owns water	17	7	0,41
3	Hurricanes	7	2	0,29
4	Birds in NYC	5	2	0,40
5	Great Lakes	4	4	1,00
6	Firearms young people	13	6	0,46
7	Gates	13	1	0,08

8	GMO Mosquitoes	17	6	0,35
9	Biden's build	25	2	0,08
10	Texas is #1	45	10	0,22
11	Reducing greenhouse emissions	14	8	0,57
12	Welfare spending	9	2	0,22
13	Fighting cap and trade	0	0	-
14	Startups in Wisconsin	14	2	0,14

Table 4.1: Overview of annotated dataset, source: Author

To provide a concrete illustration of the annotation protocol, this section presents the complete multidimensional data model for a single article from the corpus: "Cutting Emissions." The table below details each of the 23 unique data points identified within the source text. For each point, the extracted dimensions—including the Subject, Time Period, Value, Unit, and Source—are shown. Furthermore, the "Weight" assigned to each data point based on its importance to the central claim is included. This detailed view demonstrates how the abstract concept of a multidimensional fact was operationalized and serves as a specific example of the data informing the aggregated results of this study.

Metric	Region	Time period 1	Time period 2	Value	Units	Source	Cites	Weight
Carbon dioxide emissions	U.S.	Z- 15 years	Z	-14	%	Sullivan	Our World in Data	3
Rank of countries cutting greenhouse gas emissions	World	-	-	1	Order	Sullivan	Our World in Data	4
Carbon dioxide emissions	U.S.	2005	2020	-970 million	Metric tons	Sullivan	Our World in Data	4
Carbon dioxide emissions	Japan	2005	2020	-237 million	Metric tons	Sullivan	Our World in Data	3

Carbon dioxide emissions	China	2005	2020	4,689 million	Metric tons	Sullivan	Our World in Data	3
Carbon dioxide emissions	India	2005	2020	1,315 million	Metric tons	Sullivan	Our World in Data	3
Annual Carbon dioxide emissions	U.S.	2005	2019	-14	%	Author	Our World in Data	3
Annual Carbon dioxide emissions	Japan	2005	2019	-14	%	Author	Our World in Data	3
Annual Carbon dioxide emissions	U.K.	2005	2019	-35	%	Author	Our World in Data	4
Annual Carbon dioxide emissions World Ranking	U.K.	2005	2019	1	Order	Author	Our World in Data	4
Annual Carbon dioxide emissions World Ranking	Italy	2005	2019	2	Order	Author	Our World in Data	4
Annual Carbon dioxide emissions World Ranking	France	2005	2019	3	Order	Author	Our World in Data	4
Annual Carbon dioxide emissions World Ranking	Germany	2005	2019	4	Order	Author	Our World in Data	4

Annual Carbon dioxide emissions World Ranking	Japan	2005	2019	5	Order	Author	Our World in Data	4
Annual Carbon dioxide emissions World Ranking	U.S.	2005	2019	5	Order	Author	Our World in Data	4
Carbon dioxide emissions per person	U.S.	-	-	2 * Y	Metric tons	Author	Rob Jackson	4
Carbon dioxide emissions per person	China	-	-	Y	Metric tons	Author	Rob Jackson	3
Carbon dioxide emissions per person	India	-	-	8 * Y	Metric tons	Author	Rob Jackson	3
Number of people	China	-	-	1.4 billion	People	Author	Rob Jackson	3
Greenhouse gas emissions	U.S.	2005	-	879 million	Metric tons	Author	Our World in Data	3
Greenhouse gas emissions	U.S.	-	2019	607 million	Metric tons	Author	Our World in Data	3
Annual Carbon dioxide emissions	U.S.	2005	2019	14.3	%	Author	Our World in Data	3
Annual Greenhouse gas emissions	U.S.	2005	2019	9.5	%	Author	Our World in Data	4

Table 4.2: Example of multidimensional model, source: Author

Following the analysis of the source text, the same dimensional annotation process was applied to the corresponding LLM-generated summary to identify which facts were preserved. The resulting model is presented in the table below.

Of the original data points in the source article, the summary successfully captured 7 unique quantities. A qualitative inspection of this model reveals a key pattern of information loss that is central to this thesis: while the core Metric and Subject of a fact are often retained, specific quantitative Values can be abstracted into comparisons (e.g., "X" vs. "Higher >X"). Furthermore, contextual dimensions like Time Period are frequently omitted. This annotated output is used to calculate the final Coverage and Completeness scores for the article.

Metric	Subject	Time Period 1	Time Period 2	Value	Unit	Source	Cites
Rank of countries cutting greenhouse gas emissions	World	-	-	1	Order	Sullivan	unknown source
Carbon dioxide emissions	U.S.	Z- 15 years	Z	-14	%	Sullivan	Our World in Data
Greenhouse gas emissions	U.K.	-	-	-35	%	Author	Rob Jackson
Carbon dioxide emissions per person	U.S.	-	-	X	Comparison	Author	Rob Jackson
Carbon dioxide emissions per person	China	-	-	Higher >X	Comparison	Author	Rob Jackson
Carbon dioxide emissions per person	India	-	-	Much higher > Y	Comparison	Author	Rob Jackson
Annual Greenhouse gas emissions	U.S.	-	-	9.5	%	Author	Our World in Data

Table 4.3 Average of standard evaluation metrics, source: Author

4.2 Standard Evaluation Metrics

To establish a baseline for the quality of the LLM-generated summaries, standard evaluation metrics were calculated by comparing each summary to its full-text source article. Table 4.4 presents the average performance across the dataset.

BERTScore (F1)	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE - L SUM
0,855	0,263	0,123	0,167	0,233

Table 4.4: Average of standard evaluation metrics, source: Author

The results show a moderately high semantic similarity, with an average BERTScore F1 of 0.855. This indicates that, on a conceptual level, the summaries are well-aligned with the source text. In contrast, the lexical overlap is low, with an average ROUGE-L score of

0.167, which points towards a highly abstractive summarization style where the model rephrases information rather than directly extracting it.

4.3 Custom Metrics

While the baseline metrics suggest coherent summaries, the custom metrics designed for this thesis reveal a significant loss of specific, factual information. Table 4.2 presents the average scores for the three custom metrics.

Data point Coverage	Dimensional Completeness (Recall)	Weighted Data Point Coverage
0,313	0,746	0,372

Table 4.5: Average of custom evaluation metrics, source: Author

On average, the summaries captured only 31.3% of the available data points (Data Point Coverage). For the data points that were included, they retained an average of 74.6% of their original dimensional context (Dimensional Completeness). Most critically, when weighting the data points by their importance to the central claim, the summaries captured just 37.2% of the most salient information (Weighted Coverage).

The stark contrast between the high BERTScore and the low Weighted Coverage is the central finding of this study. It suggests that while the summaries are fluent and semantically related to the source, they fail to preserve the specific, dimensional evidence that underpins the fact-check's analysis. This gap is illustrated visually in Figure 1, which plots the key baseline metric against the custom metrics.

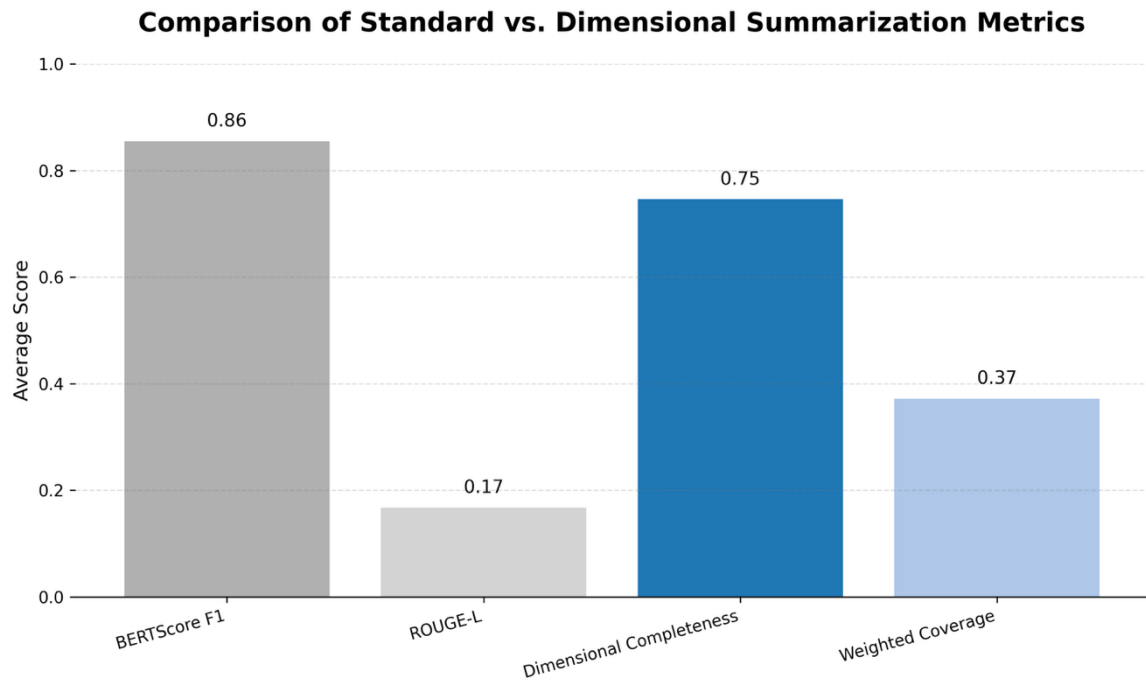


Figure 1 Comparison of Standard and Dimensional Summarization Metrics, source: Author

The distributions of these scores across the 14 articles are visualized in the box plot in Figure 2, which offers a granular view of each metric's consistency and variance. The standard metrics show stable performance: BERTScore F1 is consistently high with a median around 0.86, while ROUGE-L is consistently low, both with minimal variance. In stark contrast, the dimensional metrics exhibit significant volatility. While Dimensional Completeness has a high median (approx. 0.85), indicating that captured facts are often complete, its wide distribution and a critical outlier at 0.0 reveal a risk of severe context stripping. Most importantly, Weighted Coverage shows a low median (approx. 0.31), yet its high variance and a notable outlier near 1.0 suggest that while the model typically fails to preserve the most salient facts, achieving high factual fidelity is possible in some cases. Thus, the visualization confirms the general trend of dimensional information loss while its outliers highlight the full spectrum of the model's performance, from extreme failure to unexpected success.

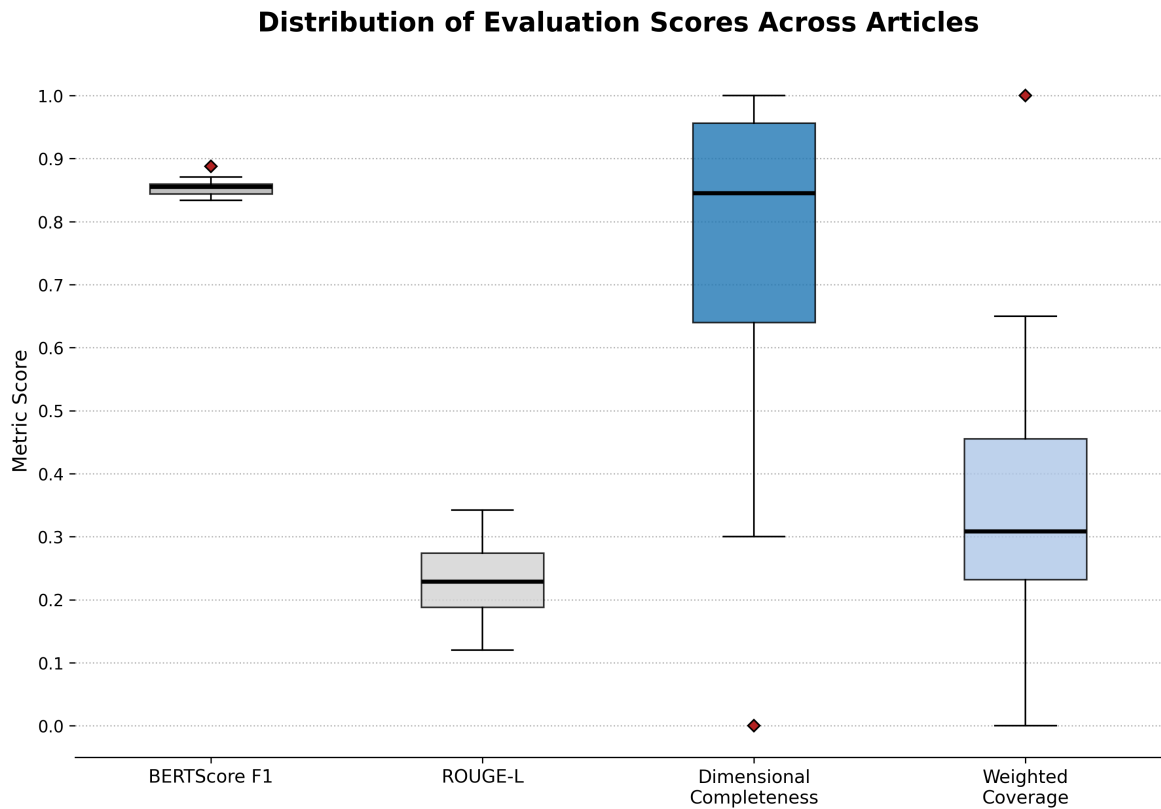


Figure 2, source: Author

4.4 Breakdown of Dimensional Recall

To understand what kind of information is being lost, the recall rate was calculated for each individual dimension type. The results, visualized in Figure 3, reveal a clear hierarchy in the preservation of contextual information.

The analysis shows a clear hierarchy in which contextual details are preserved. Fundamental dimensions that define a fact's identity—such as Subject (97%), Unit (97%), Region (92%), and even its Source (76-85%)—are captured with very high fidelity. This indicates the model is proficient at identifying the core actors and the provenance of a claim.

However, a significant drop-off occurs when preserving the specific details of the measurement itself. The recall for the exact Value is only 72%, suggesting the model often abstracts or paraphrases numerical data. Most notably, the dimension with the poorest performance is Time Period, which is correctly retained only 61% of the time.

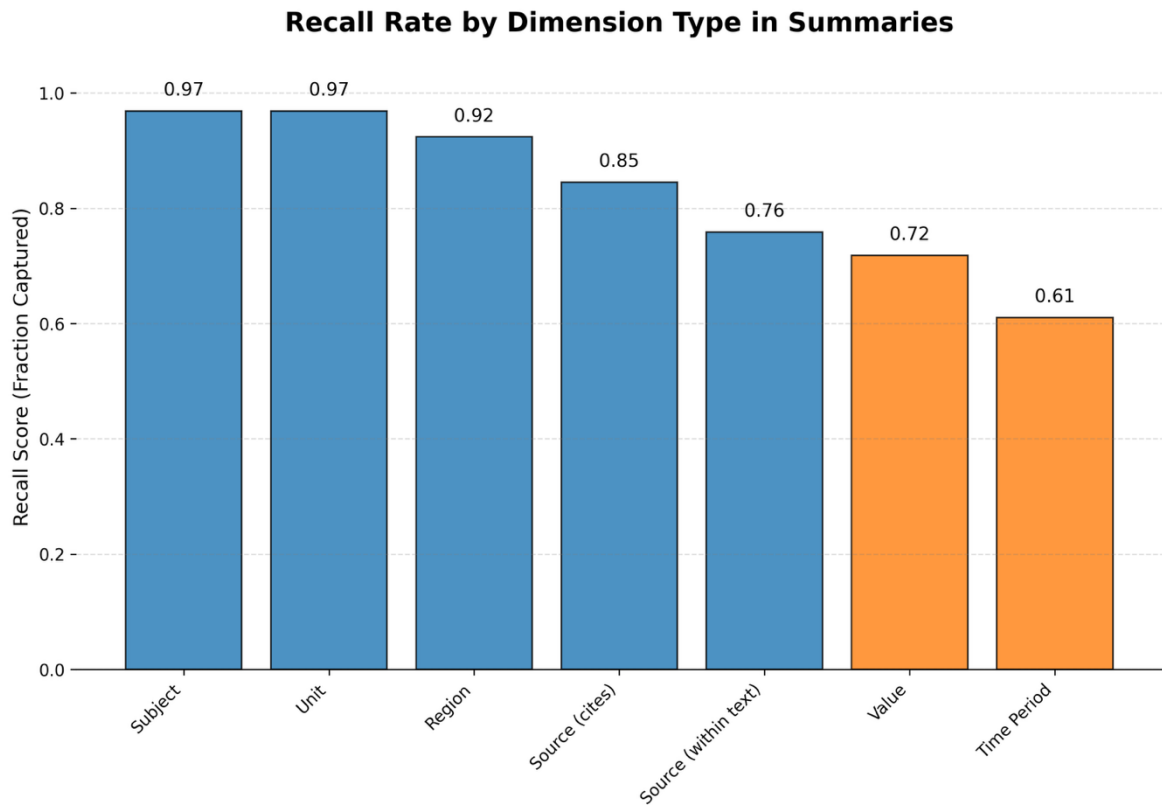


Figure 3, source: Author

This finding reveals a specific pattern of failure in the summarization process: the model successfully extracts the "what," "where," and "who" of a data point, but it frequently sacrifices the precise "how much" and, most critically, the "when." By disconnecting facts from their temporal context, the summaries risk misrepresenting claims where timing is a crucial element of their validity.

5 Discussion

This chapter interprets the findings presented in the previous chapter, discusses their implications in the broader context of AI and fact-checking, acknowledges the study's limitations, and suggests avenues for future research.

5.1 Interpretation of Key Findings

The most significant finding of this study is the blatant divergence between the summaries' performance on standard semantic metrics versus the proposed dimensional metrics. The high average BERTScore F1 (0.86) suggests that the LLM is highly proficient at paraphrasing and capturing the overall "gist" of the source text. From a coherence and topic-alignment perspective, the summaries are successful.

However, this surface-level coherence masks a critical failure in factual preservation. The low Weighted Coverage score (0.37) reveals that this fluency comes at a steep cost: the exclusion of more than 60% of the most salient data points that form the evidentiary backbone of the fact-check. This "semantic vs. factual gap" suggests that current summarization models, when given a simple prompt, are optimized for narrative flow and topical relevance rather than for dimensional integrity and the preservation of verifiable evidence.

The breakdown of dimensional recall (Figure 4.3) adds a crucial layer of nuance to this finding. The model does not lose information randomly. It demonstrates a clear hierarchy of importance, successfully preserving fundamental dimensions like Subject, Unit, and Region with over 93% recall. It is also surprisingly effective at retaining the Source of information (77-98% recall), indicating an ability to identify provenance.

The primary failure point lies in preserving the data's specific temporal and quantitative context. The recall for the exact Value is only 72%, often due to the model abstracting specific numbers (e.g., "\$555 Billion") into vague descriptions (e.g., "a substantial fraction"). Most notably, the dimension with the poorest performance is Time Period, which is correctly retained only 62% of the time. This reveals a specific pattern of failure: the model successfully extracts the "what," "where," and "who" of a data point, but it frequently sacrifices the precise "how much" and, most critically, the "when." By disconnecting facts from their temporal context, the summaries risk misrepresenting claims where timing is a crucial element of their validity.

5.2 Robustness of the Weighted Coverage Metric

To ensure the Weighted Coverage finding was not an artifact of the specific 1-5 linear weighting scale, a sensitivity analysis was performed. The analysis (presented in Table 5.1)

confirms that the conclusion holds across linear, exponential, and diminishing-returns-based weighting schemes, which strengthens the validity of the central finding.

Weighting Scheme	Resulting Weighted Coverage Score
Squared Weights	0,413
Linear (Original)	0,372
Diminishing Returns	0,315

Furthermore, the pattern observed in the sensitivity analysis reveals that the LLM is more likely to capture the most critical, high-weight data points (as shown by the higher score in the "Squared Weights" scheme) at the expense of supplementary context. This confirms the trade-off between capturing the core claim and retaining its surrounding dimensional information.

5.3 Implications of the Study

The findings of this study have direct implications for both the developers and consumers of AI-driven text summarization.

- For AI Developers and Researchers: This work highlights the inadequacy of standard metrics like ROUGE and BERTScore for evaluating summaries in high-stakes, factual domains. It demonstrates the need to develop and optimize summarization models not just for semantic similarity, but for the explicit preservation of structured, factual data and their dimensional attributes. The proposed metrics offer a starting point for such dimension-aware evaluation.
- For Fact-Checking Organizations and Journalists: Automated summaries, in their current form, should not be used as direct replacements for human-written abstracts without rigorous verification. The tendency of LLMs to drop temporal and quantitative details means that a generated summary could inadvertently misrepresent the findings of a fact-check. The methodology proposed here could serve as a blueprint for a "verification tool" to help editors quickly assess the factual completeness of a generated summary.

5.4 Limitations of the Study

It is important to acknowledge the limitations of this exploratory study, which define the boundaries of its conclusions.

The primary limitation is the single-annotator methodology for both dimensional modeling and weight assignment. While the process was guided by systematic rules, it is subject to individual interpretation. A more robust study would require multiple annotators and a calculation of inter-annotator agreement to validate the annotation scheme.

The small dataset size ($N=14$) and its focus on a single source (PolitiFact) and domain (climate/environment) mean the results should be considered an in-depth case study. The findings may not be generalizable to all fact-checking outlets or subject areas.

The use of a single LLM (gpt-4o-mini) with a single, simple prompt provides a valuable baseline but does not explore the full potential of current technology. Performance could vary significantly with different models or more advanced prompting strategies (e.g., chain-of-thought or few-shot prompting).

5.5 Future Work

This study opens several avenues for future research. The most immediate step is to address the limitations by validating the dimensional annotation scheme with multiple annotators to establish its reliability. Subsequently, the evaluation framework should be applied to a larger, more diverse dataset and a wider range of LLMs and prompting techniques to test the generalizability of these findings. Finally, the ultimate goal would be to use this dimensional evaluation framework not just for analysis, but as a reward signal for fine-tuning a specialized summarization model that is explicitly optimized for factual and dimensional completeness.

6 Conclusion

The proliferation of information and misinformation has elevated the importance of fact-checking journalism. As the volume of fact-checking content grows, so does the need for effective summarization to make these detailed reports accessible. This thesis addressed the critical challenge of ensuring that summaries, particularly those generated by Large Language Models, preserve the essential factual evidence of the source text.

The core contribution of this work is the proposal and application of a novel, dimension-aware evaluation framework. By conceptualizing factual data points as multidimensional objects—defined by attributes such as time, region, and source—this study moved beyond standard semantic metrics to quantify the preservation of evidentiary context.

Applying this framework to a set of 14 fact-checking articles summarized by a state-of-the-art LLM yielded a clear and significant finding: while automated summaries are semantically coherent, they consistently fail to capture a large portion of the crucial, dimensional data that underpins a fact-check's conclusion. Specifically, the analysis revealed a pattern where the model effectively retains the subject and source of a fact but struggles to preserve its precise value and, most critically, its temporal context.

This work demonstrates that for high-stakes domains like fact-checking, fluency is not a sufficient measure of a summary's quality. The informational integrity of a summary is contingent upon the preservation of its dimensional grounding. By providing a method to measure this, this thesis offers a new lens for evaluating the true utility of automated summarization and provides a concrete path forward for developing AI systems that are not only fluent but also factually reliable.

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Appendices

Appendix A: Quantity Extraction Script

```
"""
Quantity Extraction Script for Fact-Checking Articles.

This script uses the spaCy NLP library to identify and extract numerical quantities
and their associated units from text. It is designed to be a semi-automated first pass
for the dimensional modeling process described in this thesis.

Dependencies:
- spacy>=3.0
- A spaCy English model (e.g., en_core_web_sm or en_core_web_lg)

Setup:
1. Install dependencies: pip install spacy
2. Download a model: python -m spacy download en_core_web_sm
"""

import spacy
from spacy.matcher import Matcher
from typing import List, Dict, Set, Tuple

# --- 1. Lexicon and Model Initialization ---

# Load the spaCy model
try:
    nlp = spacy.load("en_core_web_sm")
except OSError:
    print("spaCy model not found. Please run: python -m spacy download en_core_web_sm")
    exit()

# A comprehensive lexicon of units, all in lowercase for consistent matching.
UNIT_LEXICON: Set[str] = {
    # Percentages
    "%", "percent", "percentage", "percentage points", "points",
    # Time
    "second", "seconds", "min", "mins", "minute", "minutes",
    "hour", "hours", "day", "days", "week", "weeks", "month", "months",
    "year", "years", "yr", "yrs", "decade", "decades", "century", "centuries",
    # Counts/Magnitudes
```

```

"million", "billion", "trillion", "thousand", "hundred",
# Currencies
"usd", "dollar", "dollars", "€", "euro", "euros", "eur", "£", "pound", "pounds",
# Physical Measurements
"meter", "meters", "m", "kilometer", "kilometers", "km", "mile", "miles", "mi",
"foot", "feet", "ft", "acre", "acres", "hectare", "hectares",
"gallon", "gallons", "liter", "liters", "l", "acre-foot", "acre-feet",
"kg", "kilogram", "kilograms", "ton", "tons", "tonne", "tonnes", "lb", "lbs",
# Ordinal/Categorical
"category", "rank"
}

# --- 2. Matcher Setup ---

def setup_matcher(vocab) -> Matcher:
    """Initializes the spaCy Matcher with custom patterns."""
    matcher = Matcher(vocab)

    # Pattern 1: A number followed immediately by a unit from the lexicon.
    # e.g., "100 acres", "1.5 million", "50 %"
    pattern_num_unit = [
        {"LIKE_NUM": True},
        {"LOWER": {"IN": UNIT_LEXICON}}
    ]
    matcher.add("NUMBER_UNIT", [pattern_num_unit])

    return matcher

matcher = setup_matcher(nlp.vocab)

# --- 3. Main Extraction Function ---

def extract_quantities(text: str) -> List[str]:
    """
    Extracts number-unit pairs and standalone numerical entities from a given text.

    This function prioritizes spaCy's built-in Named Entity Recognition (NER) for
    standard types like PERCENT and MONEY, then uses a custom Matcher to find
    other number-unit combinations, avoiding double-counting.

    Args:
        text: The input string to process.

    Returns:
        A list of unique quantity strings found in the text.
    """

```

```

"""
if not isinstance(text, str) or not text.strip():
    return []

doc = nlp(text)
found_metrics: Set[str] = set()
handled_tokens: Set[int] = set()

# Step 1: Prioritize spaCy's built-in entity recognizer for high-precision entities.
for ent in doc.ents:
    if ent.label_ in ["PERCENT", "MONEY", "QUANTITY", "CARDINAL"]:
        found_metrics.add(ent.text.strip())
        for token in ent:
            handled_tokens.add(token.i)

# Step 2: Use the custom rule-based matcher for other combinations.
matches = matcher(doc)
for match_id, start, end in matches:
    # Check if this match overlaps with an already handled entity.
    if any(token_idx in handled_tokens for token_idx in range(start, end)):
        continue

    span = doc[start:end]
    found_metrics.add(span.text.strip())

    # Add all tokens in this match to the handled set to prevent future overlaps.
    for i in range(start, end):
        handled_tokens.add(i)

return sorted(list(found_metrics))

# --- 4. Example Usage ---

if __name__ == '__main__':
    # This block runs only when the script is executed directly.
    # It serves as a demonstration and a simple test case.

    example_text = """
    According to the CBO, the official legislated cost of the bill is 1.75 Trillion USD
    over 10 years. However, the Committee for a Responsible Federal Budget estimated that
    the true cost could be almost $5 trillion if the provisions were made permanent.
    The bill expands welfare programs by the largest amount in 60 years and includes
    over 555 Billion USD for climate change provisions, which is nearly-third of the
    total.

    The first-year increase in spending is projected to be 167 Billion USD, with an

```

```

unemployment rate of around 4 percent. This represents a 0.9% share of the economy.
"""

print("--- Example Text ---")
print(example_text)

extracted_data = extract_quantities(example_text)

print("\n--- Extracted Quantities ---")
if extracted_data:
    for metric in extracted_data:
        print(f"- {metric}")
else:
    print("No quantities found.")

```

Appendix B: Evaluation Metrics

ID	Articles	BERT Precision	BERT Recall	BERT F1	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-L sum
1	Cutting Emissions	0,873	0,841	0,856	0,285	0,128	0,165	0,257
2	Billionaire couple owns water	0,878	0,830	0,853	0,228	0,124	0,149	0,204
3	Hurricanes	0,880	0,829	0,854	0,313	0,147	0,196	0,278
4	Birds in NYC	0,914	0,864	0,888	0,368	0,162	0,237	0,342
5	Great Lakes	0,901	0,842	0,871	0,353	0,210	0,275	0,319
6	Firearms young people	0,899	0,839	0,868	0,353	0,184	0,223	0,306
7	Gates	0,893	0,827	0,859	0,246	0,115	0,156	0,223
8	GMO Mosquitoes	0,858	0,811	0,834	0,212	0,090	0,116	0,187
9	Biden's build	0,866	0,846	0,856	0,200	0,086	0,123	0,182
10	Texas is #1	0,853	0,829	0,841	0,133	0,062	0,087	0,120
11	Reducing greenhouse emissions	0,864	0,819	0,841	0,189	0,080	0,103	0,157

12	Welfare spending	0,868	0,828	0,847	0,230	0,080	0,133	0,191
13	Fighting cap and trade	0,861	0,825	0,843	0,263	0,131	0,181	0,234
14	Startups in Wisconsin	0,889	0,833	0,860	0,315	0,128	0,198	0,261

Appendix D: Custom Metrics

ID	Articles	Coverage	Recall	Weighted Coverage
1	Cutting Emissions	0,348	0,833	0,313
2	Billionaire couple owns water	0,273	0,530	0,529
3	Hurricanes	0,286	0,900	0,417
4	Birds in NYC	0,600	0,950	0,650
5	Great Lakes	1,000	1,000	1,000
6	Firearms young people	0,462	0,806	0,468
7	Gates	0,200	1,000	0,250
8	GMO Mosquitoes	0,118	0,300	0,200
9	Biden's build	0,160	0,958	0,247
10	Texas is #1	0,178	0,685	0,304
11	Reducing greenhouse emissions	0,143	1,000	0,227
12	Welfare spending	0,333	0,625	0,214
13	Fighting cap and trade	0,000	0,000	0,000
14	Startups in Wisconsin	0,286	0,857	0,386

Appendix D: Sensitivity Analysis of Weight Scaling

ID	Articles	Weighted Coverage	Weighted Coverage Squared	Weighted Coverage Root
1	Cutting Emissions	0,313	0,320	0,308
2	Billionaire couple owns water	0,529	0,207	0,139
3	Hurricanes	0,417	0,556	0,349
4	Birds in NYC	0,650	0,702	0,624
5	Great Lakes	1,000	1,000	1,000
6	Firearms young people	0,468	0,470	0,466
7	Gates	0,250	0,340	0,090
8	GMO Mosquitoes	0,200	0,283	0,158
9	Biden's build	0,247	0,351	0,201
10	Texas is #1	0,304	0,469	0,235
11	Reducing greenhouse emissions	0,227	0,325	0,183
12	Welfare spending	0,214	0,333	0,333
13	Fighting cap and trade	0,000	0,000	0,000
14	Startups in Wisconsin	0,386	0,429	0,319